

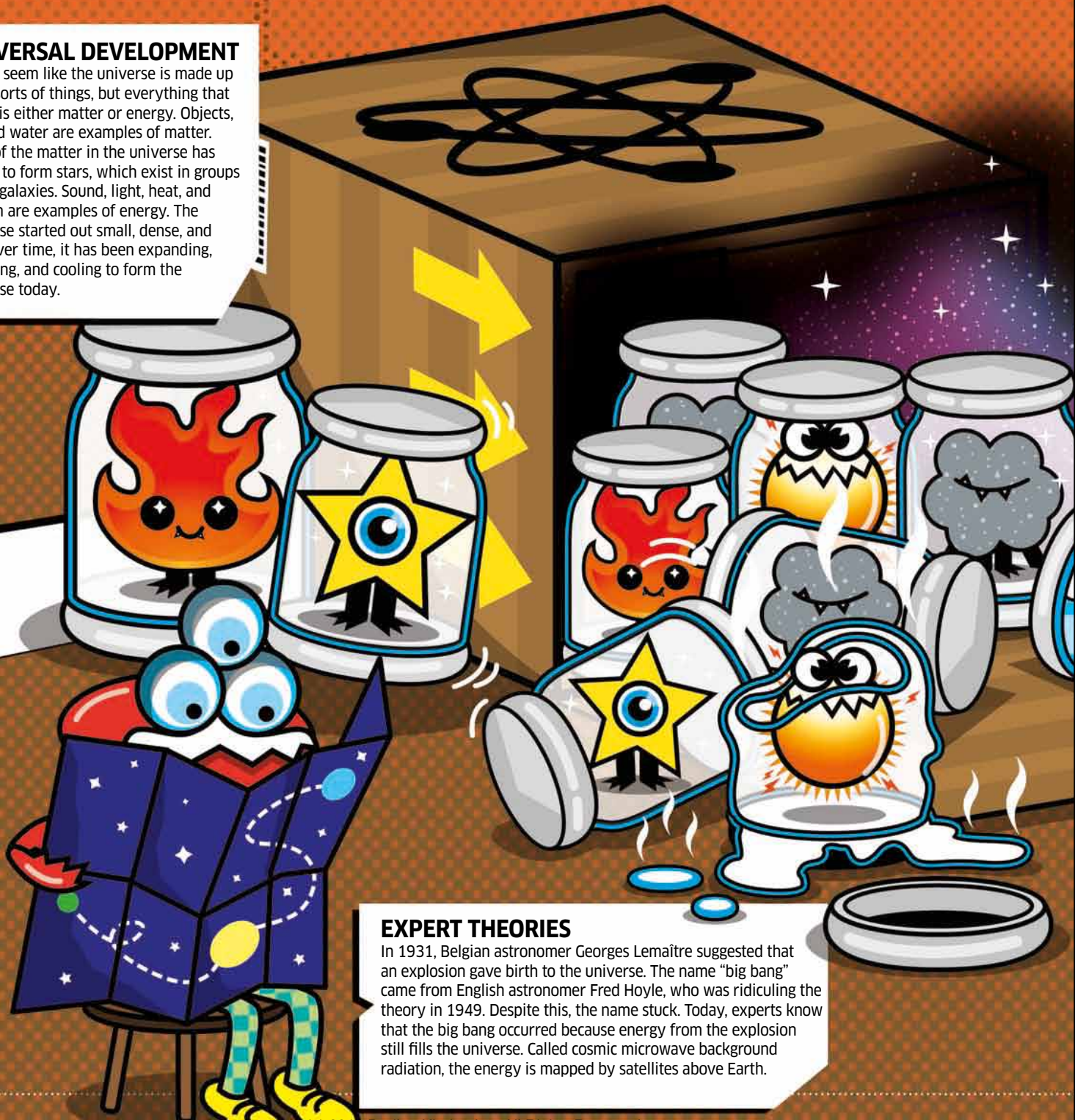
# WHERE DID THE UNIVERSE COME FROM?

**BANG!** That's how the universe started. A huge explosion, now known as the big bang, marked the beginning of everything, about 13.7 billion years ago. Scientists believe that in that moment, the entire universe, including time and space, got under way. It was even smaller than a period back then and was made up of miniscule particles of energy. It's difficult to comprehend, and even scientists don't know exactly why the big bang occurred, but they agree that this is where the universe had its beginning.

## ALL ROUTES

### UNIVERSAL DEVELOPMENT

It may seem like the universe is made up of all sorts of things, but everything that exists is either matter or energy. Objects, air, and water are examples of matter. Most of the matter in the universe has joined to form stars, which exist in groups called galaxies. Sound, light, heat, and motion are examples of energy. The universe started out small, dense, and hot. Over time, it has been expanding, changing, and cooling to form the universe today.



### EXPERT THEORIES

In 1931, Belgian astronomer Georges Lemaître suggested that an explosion gave birth to the universe. The name "big bang" came from English astronomer Fred Hoyle, who was ridiculing the theory in 1949. Despite this, the name stuck. Today, experts know that the big bang occurred because energy from the explosion still fills the universe. Called cosmic microwave background radiation, the energy is mapped by satellites above Earth.